

Data Vortex - Aaruush '26

Round 1 Phase 2 - Logic Explanation

Team: Punch Queries **Members:** Sahil Patel, Suhani Gupta

Selected questions: E1, M5, H3

E1 - Platform Popularity

Determine which social media platform has the highest number of posts. Ignore posts where the platform is missing. Return the platform name and number of posts.

Objective

Find the platform with the highest post count while ignoring missing platform values.

Data used

dbo.posts. The cleaned model represents missing source platform values as 'Unknown'.

Logic

1. Filter out platform = 'Unknown'. 2. Group remaining posts by platform. 3. Count rows for each platform. 4. Sort counts descending. 5. Return the highest-count platform with TOP (1). A secondary platform sort makes the result deterministic if counts tie.

Why correct

The questionnaire asks for platform and number of posts and explicitly says to ignore missing platform values.

Output

platform, post_count

M5 - Detect Suspicious Engagement

Find posts where the number of shares is greater than the number of likes and comments combined. Return the top 20.

Objective

Find posts where shares are greater than likes plus comments, then return the top 20.

Data used

dbo.posts. likes, shares, comments, and the persisted computed engagement column are available in the final schema.

Logic

1. Evaluate the rule shares > likes + comments for every post. 2. Keep only rows satisfying that condition. 3. Sort by total engagement descending so the highest-impact suspicious posts appear first. 4. Use post_id as a deterministic tie-breaker. 5. Return TOP (20).

Schema note

The repository's staging table contains a shares_exceed_likes flag, but the final dbo.posts schema does not persist that flag. Therefore the submitted query calculates the condition directly from the base metrics. This keeps the SQL aligned with the questionnaire and the final schema.

Output

post_id, platform, likes, shares, comments, engagement

H3 - Platform Performance Compared With Its Own Average

For every platform, identify posts whose engagement is significantly higher than the average engagement of that platform. A post is exceptional if its engagement is at least 2x the average engagement of its platform.

Objective

Find posts whose engagement is at least two times the average engagement of their own platform.

Data used

dbo.posts. engagement is defined by the repository schema as likes + shares + comments and is persisted as a computed column.

Logic

1. Use a CTE to create a scoring layer. 2. Calculate AVG(engagement) OVER (PARTITION BY platform), giving every post the average engagement of its own platform. 3. Compare each post's engagement with 2 x its platform average. 4. Keep rows meeting engagement $\geq 2 \times$ platform average. 5. Return the post metrics, platform average, and exceptional threshold for transparent verification.

Why window function

PARTITION BY platform makes the benchmark platform-specific. A single global average would incorrectly compare platforms with different engagement levels.

Output

post_id, user_id, platform, likes, shares, comments, engagement, platform_avg_engagement, exceptional_threshold

SSMS screenshot naming

Capture each query result in SSMS with the query text visible in the same frame, matching the repository submission checklist.

Question	JPEG filename
E1	01_E1_Platform_Popularity_SSMS.jpeg
M5	02_M5_Suspicious_Engagement_SSMS.jpeg
H3	03_H3_Platform_Performance_SSMS.jpeg

Submission check: Verify all three outputs in SSMS before capturing screenshots. Do not manually type result values into the report; use actual query output.